

Title: Is Health Impact Based on the County's Food Accessibility

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Summary of research questions:

1. Does the level of rural or urban a county is affect the health outcomes of its population?

First, this research question will allow us to understand if a rural or urban county will affect the health of those specific populations, whether living in one county may benefit or harm their health.

2. How does a county's accessibility to food affect the health outcomes of its population?

Next, this research question targets the effect of a county's accessibility to food and the health outcome it will have on the population. As food accessibility is a big factor in health for specific populations, comparing the two datasets will show the increase or decrease in health based on the county's food accessibility.

3. Does the geographical location of each county affect health outcomes and food access?

Lastly, this research question will inform us whether a county's geographical location shows any patterns in residents' access to food and their health outcomes. If a specific region shows a common occurrence of a specific food accessibility issue or higher rates of a health outcome, we can see the connection between the variables.

Motivation:

The motivation for our project came from the idea that lower-population areas of the United States of America may have poorer health outcomes than areas with higher populations, due to factors such as food accessibility, isolation, and rural lifestyles. With this question in mind, our team focused on understanding the health impacts that population and food accessibility have on the health of people in rural or urban continuums.

Datasets:

PLACES: Local Data for Better Health, County Data, 2025 release

https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-County-Data-20/swc5-untb/about_data

Food Environment Atlas:

<https://www.ers.usda.gov/data-products/food-environment-atlas/data-access-and-documentation-downloads>

Rural-Urban Continuum Codes:

<https://www.ers.usda.gov/data-products/rural-urban-continuum-codes>

Challenge goals:

1. Multiple Datasets

For this project, we plan on including the data from three different sources to create a more in-depth analysis and fully explore our research questions. We plan on joining the datasets together by county and exploring potential patterns and relationships between the different variables.

2. New Library

For this project, we plan to use the Plotly library to create choropleth maps that visualize geographical patterns in health and food indicators. This will help us identify regional trends that may be harder to recognize without using a county-based map.

Method:

We had initially planned to merge all three of the datasets together to make one large one that we could make our visualizations from. However, due to differences between the years the data was taken, we made two merged datasets, each with two datasets. For the first merged dataset, we combined the original Places Dataset from 2025 and the Rural Urban Continuum codes from 2023 together. Although the Places Dataset was from 2025, the data inside was taken in 2023. For the second merged dataset, we combine a new Places Dataset from 2023 and the Food Environment Atlas. The Places dataset from 2023 has data taken from 2020, which matches the Food Environment Atlas.

1. Does the level of rural or urban a county is affect the health outcomes of its population?

For the first research question, we plan on exploring whether there is a connection between a county's rural or urban classification and differences in health outcomes. For this question, we will use the Rural-Urban Continuum Codes and the PLACES County Data. The Rural-Urban Continuum Code contains classifications from 1-9, where 1 is the most urban and 9 is the most rural. We plan on grouping counties by this code and calculating the summary statistics for each group for the specific health outcomes, like diabetes and obesity, and then plotting them each on a separate box plot. This will allow us to examine how the spread of the data differs between different groups, allowing us to see whether a region being more rural or urban has an effect on the health of its residents.

Boxplots (one for each indicator, and the spread will be the percentage)

X - rural/urban indicators

Y - percentage of different health outcomes

2. How does a county's accessibility to food affect the health outcomes of its population?

For this research question, we plan on exploring whether there's a connection between the accessibility of food for a county's residents and important health outcomes. We planned on using the Food Environment Atlas to calculate the ratio between fast food restaurants and grocery stores for each county, and see if there is any relationship between the ratio and different health outcomes. However, due to the number of incomplete values, we pivoted to using the number of grocery stores per 1,000 residents in each county. We created separate scatterplots for each health outcome with the number of grocery stores per 1,000 residents on the x-axis and the outcome percentage on the y-axis. If counties with a higher number tend to have higher percentages of negative health outcomes, this may show a relationship between food access and population health.

Scatterplots

X - Number of grocery stores per 1,000 residents

Y - percentage of different health outcomes

3. Does the geographical location of each county affect health outcomes and food access?

For this research question, we plan on using county-based maps to explore whether there are any geographical patterns in health and food access. For food access, we will use the Food Environment Atlas to create a choropleth map where each county is colored based on the ratio of fast food restaurants to grocery stores. We also plan on creating an additional choropleth map of health outcomes using the PLACES County data. These maps will allow us to visually identify whether certain regions have higher rates of negative health outcomes or less food access.

Geographical map of the United States based on the counties

X - percentage of different health outcomes, Number of grocery stores per 1,000 residents

EDA Results:

How large is your dataset?

Dataset (PLACES: Local Data for Better Health, County Data, 2025 release):

The rows in this dataset represent 40 different measures, as there are around 125,000-126,000 rows of data, each row capturing a geographic unit. There are 3,143 U.S. counties and similar geographic units, and this is all measured and repeated in cases, as the rows in this dataset. The columns in this dataset contain fields such as year, state abbreviations, data value estimates, confidence limits, and data value units. There are 22 different columns in this dataset. The dataset has 40 measures for 12 health outcomes, 7 prevention services, 4 diseases, 7 for disability cases, 3 for health statuses, and 7 for social health needs.

Food Environment Atlas:

The rows in this data set each represent a U.S. county, having 3,143 rows, as there are 3,143 counties in the U.S. The dataset's columns contain 300 variables, each with different FIPS codes, state abbreviations, and county names. The three main categories that the dataset has are Food Choices, showing the proximity of healthy and affordable food, health and well-being, focusing on ideas such as food insecurity, diabetes, and obesity, and community awareness, which takes into account demographic, income, and status.

Rural-Urban Continuum Codes:

The rows in the dataset represent the different counties and equivalents. The territories are assigned codes, and the area that the counties are a part of shows the size of the urban area. The dataset's columns are: FIPS, State, County_Name, Attribute, and Value. Each of these column categories shows the urban continuum codes and how we can use them to help our project.

Do your datasets have any missing data?

The Places dataset has minimal missing data, as only 66 rows, which is less than 0.03% of the entire dataset, have missing data. The columns are mostly empty in these 66 rows and are marked with a note that says "Estimates suppressed for population less than 50." The Food Environment Atlas dataset has many more missing values. We had originally planned on taking a ratio between the number of fast food restaurants and the number of grocery stores in a county. However, due to more than a third of the values being missing between the two values, we decided to focus solely on the number of grocery stores per

1,000 residents in a county. This still leaves around 900 counties without a value; however, although the Food Environment Atlas contains data from 2016, which had significantly lower numbers of missing values, the Places dataset for 2016 isn't based on counties but rather cities. The Rural-Urban Continuum Codes have no missing data and have all 9,703 rows filled out.

What are the variables of interest for your research questions?

The independent variable in the Rural-Urban Continuum Codes is RUCC_2023, which classifies each county's rurality on a scale between 1 and 9. For the Food Environment Atlas, we focused on the number of grocery stores in a county per 1,000 residents. In the PLACES dataset, the variables we focused on were the percentage of residents in a county who had diabetes and obesity

Give a summary of each variable of interest.

Rural-Urban Continuum Codes

RUCC_2023 code	Description	County Count
1	Metro $\geq$ 1 million population	483
2	Metro 250,000 to 1 million	398
3	Metro $<$ 250,000	371
4	Nonmetro $\geq$ 20,000 adjacent to metro	204
5	Nonmetro $\geq$ 20,000 not adjacent	81
6	Nonmetro 5,000 - 20,000 adjacent	385
7	Nonmetro 5,000 - 20,000 not adjacent	248
8	Nonmetro $<$ 5,000 adjacent	468
9	Nonmetro $<$ 5,000 not adjacent	595

PLACES 2025 Dataset

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max
Diabetes	2957	13.64	2.78	4.90	11.80	13.30	15.20	27.10
Obesity	2957	37.42	4.75	16.70	34.90	37.90	40.40	52.90

Places 2020 Dataset

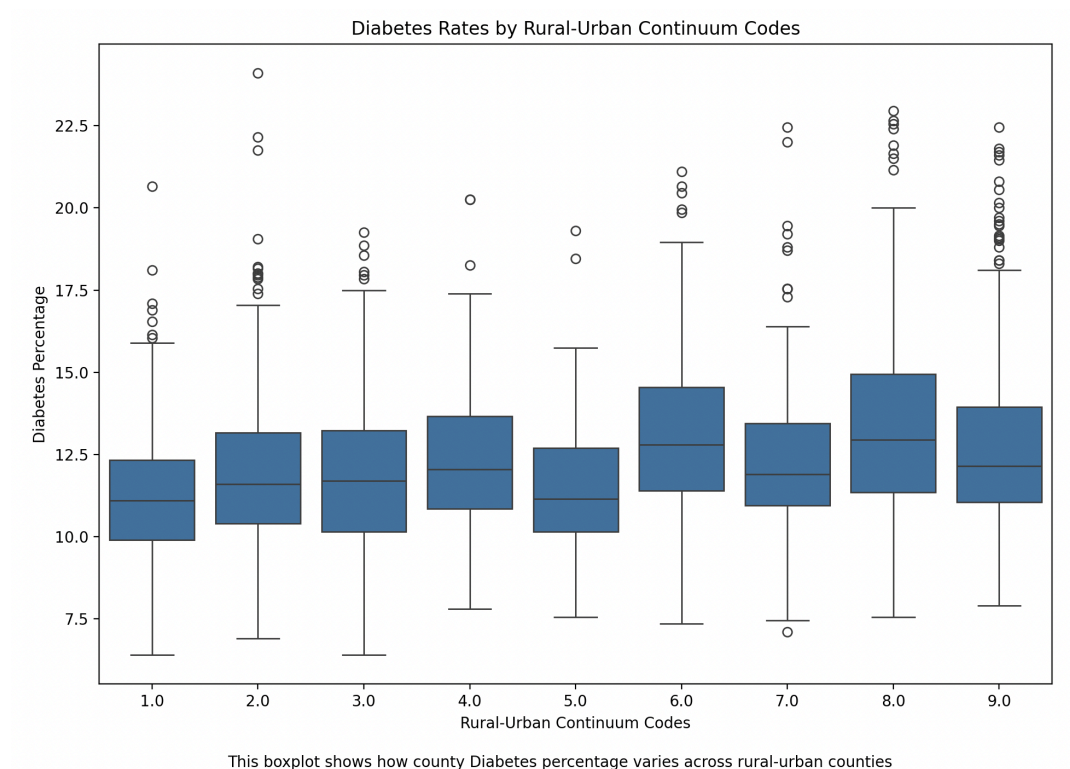
Variable	n	Mean	SD	Min	Q1	Median	Q3	Max
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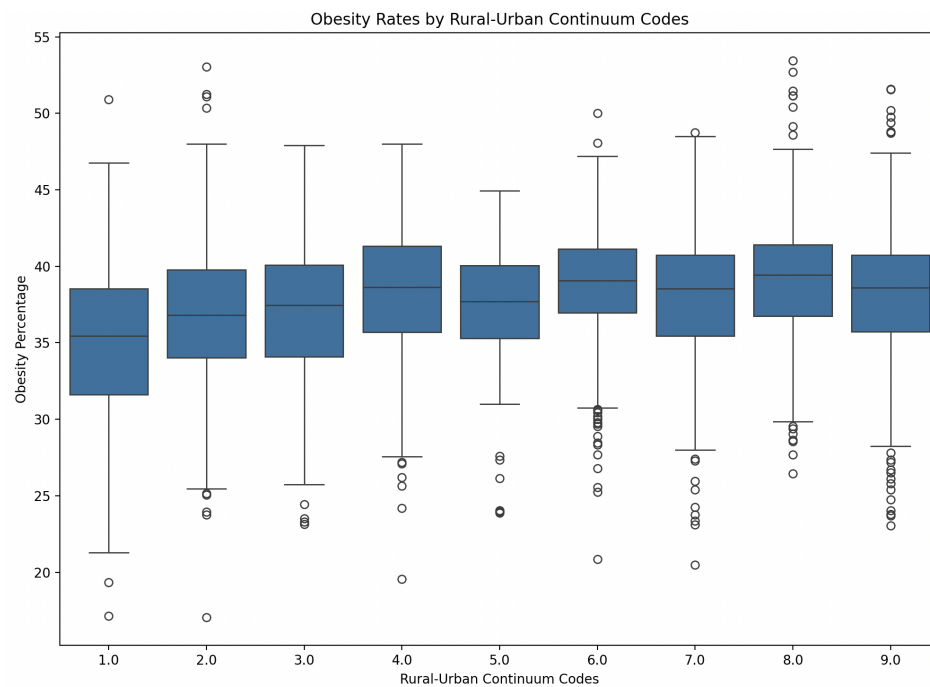
Diabetes	2176	11.43	2.25	5.9	9.85	11.15	12.65	21.75
Obesity	2176	36.87	4.65	17.15	34.35	37.25	39.95	50.8

Food Environment Atlas

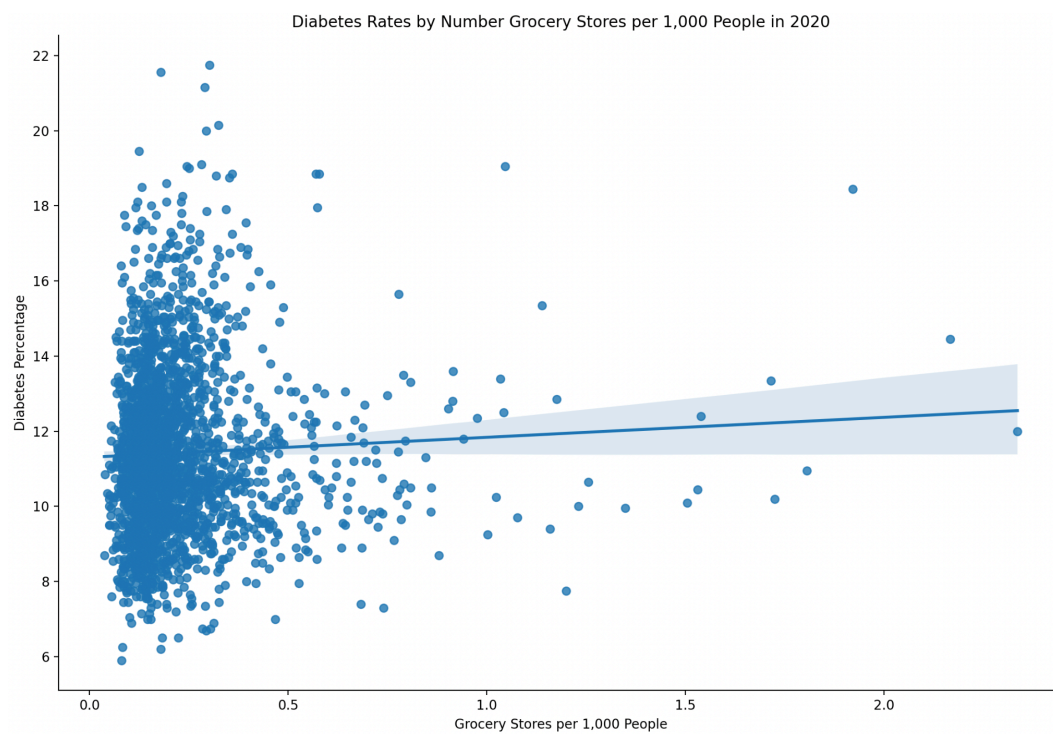
Variable	Description	n	Mean	SD	Min	Q1	Median	Q3	Max
GROCPTH20	The number of grocery stores per 1,000 people	2176	0.227	0.18	0.037	0.134	0.18	0.257	2.335

Visualizations and Results:

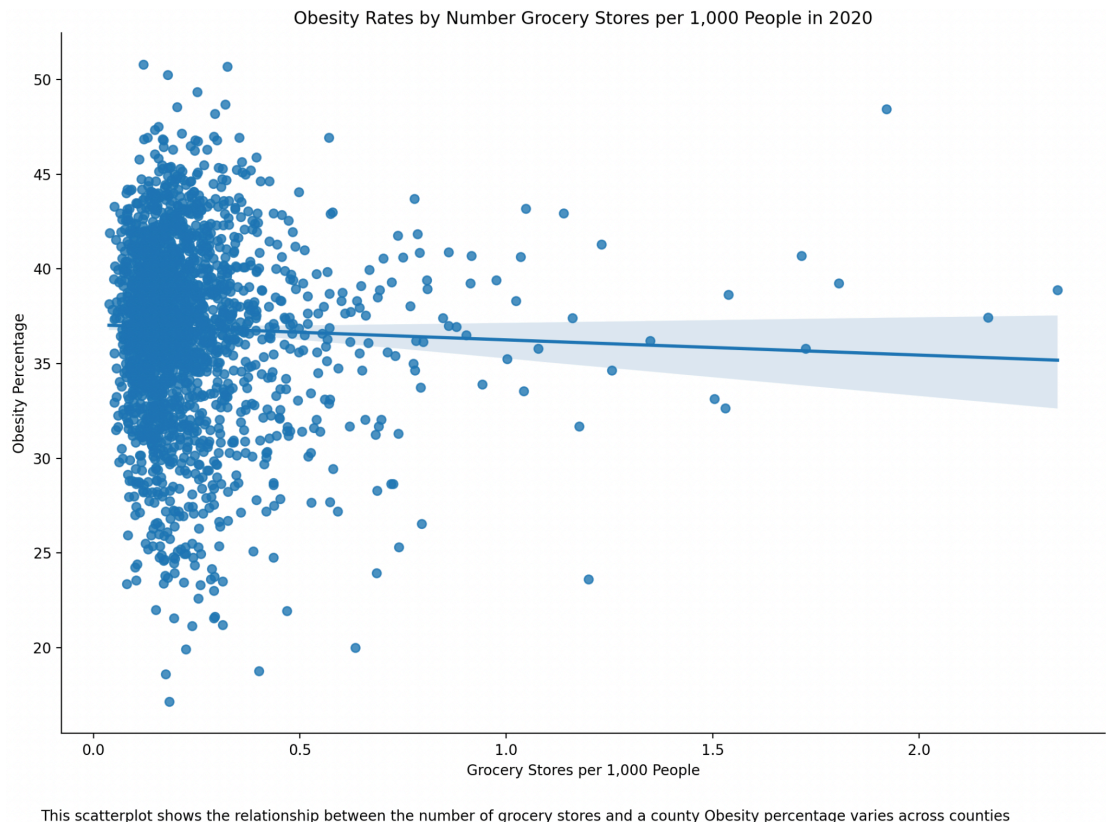




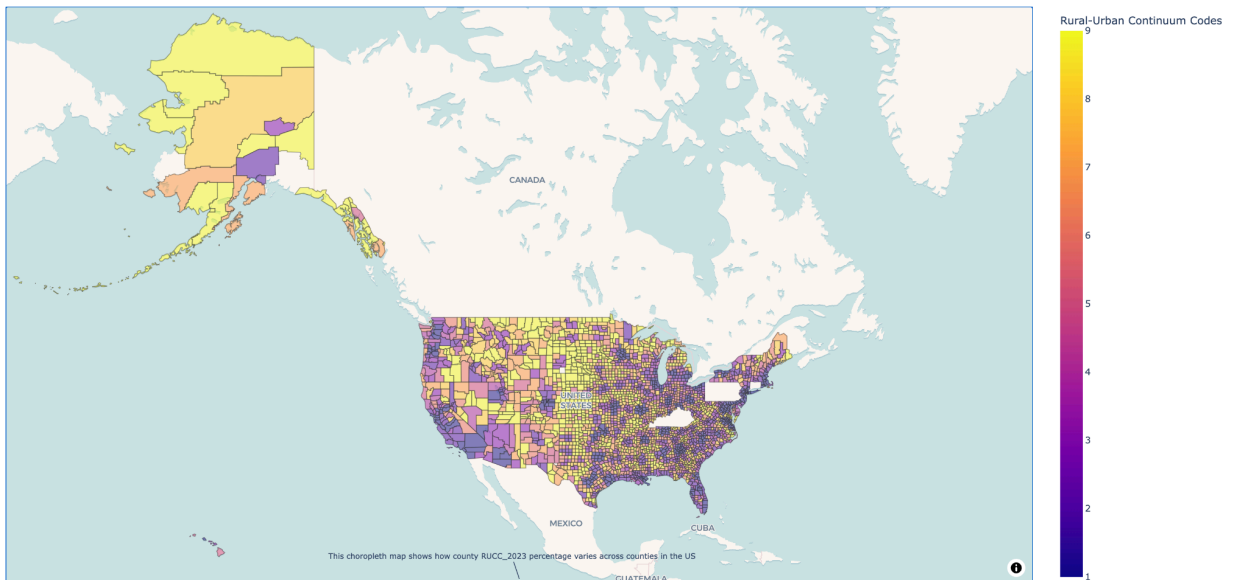
This boxplot shows how county Obesity percentage varies across rural-urban counties



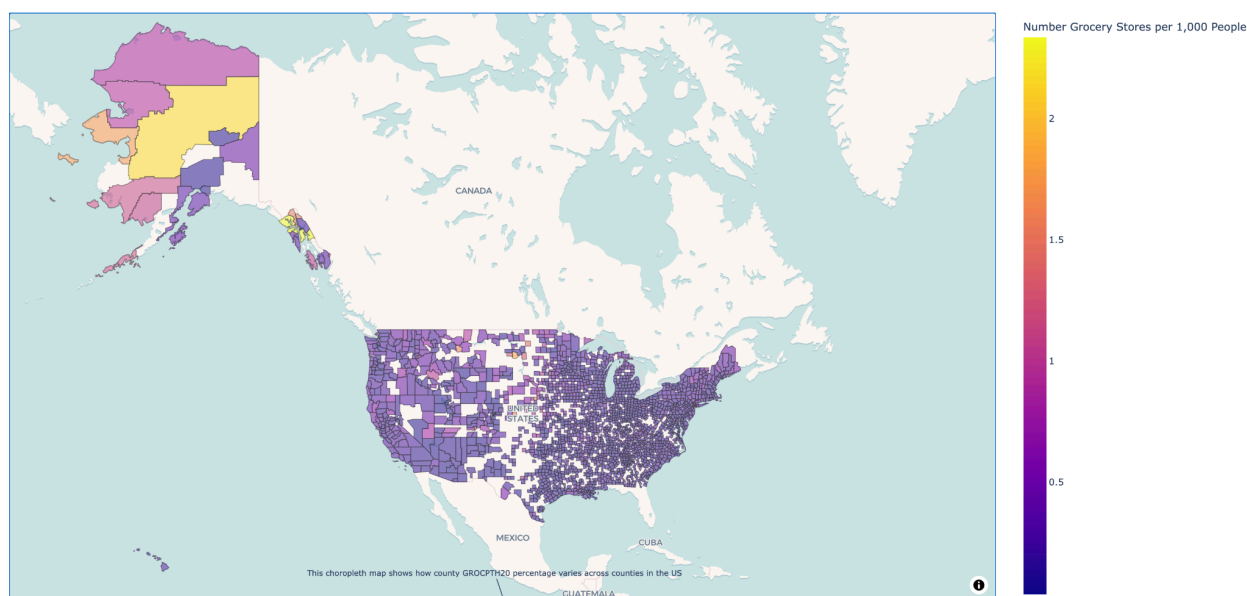
This scatterplot shows the relationship between the number of grocery stores and a county Diabetes percentage varies across counties



Rural-Urban Continuum Codes by County



Number Grocery Stores per 1,000 People by County



Multiple Dataset Summary:

All three datasets use the FIPS code as a common key, allowing them to be joined. RUCC is used as the classification, the Food Atlas provides food access indicators, and PLACES provides the health outcomes for each county.

Results:

Research Question 1: Does the level of rural or urban a county is affect the health outcomes of its population?

The boxplots that compared the diabetes and obesity rates across the Rural-Urban Continuum Codes revealed a visible trend, which was that counties that were classified as more rural (codes 6-9) had a slightly higher median diabetes rate compared to more urban counties (codes 1-3), suggesting that the rurality may be associated with worse health outcomes. Regarding obesity rates, the pattern is similar to the diabetes rate; however, it is less emphasized, with median obesity rates appearing consistently across RUCC codes, while the spread and upper outliers increased in rural areas. One interesting finding in the urban counties (code 1) was that it displayed the widest downward variance in both diabetes and obesity rates, showing that large metro areas contain healthy counties but also have notable outliers near the higher end. This idea shows the reflection of large cities and how wealthy urban populations bring down the median, while low-income urban communities cause the upper tail to increase. These results suggest that rurality does appear to be associated with poorer health outcomes; however, the causation is not strictly linear, and urbanization does not solely guarantee better population health.

Research Question 2: How does a county's accessibility to food affect the health outcomes of its population?

The scatterplots of grocery stores per 1,000 residents against diabetes and obesity rates revealed a relationship that was weak, but had importance. Counties with low grocery store concentration (clustered near 0.0-0.3 stores per 1,000 people) showed a high variability in health outcomes, with some of the

highest rates of diabetes and obesity concentrated in this significant range. As grocery store density increases, the number of health outcomes narrows rather than spreads, and the trend line for diabetes shows a small positive slope, which was unexpected. The grocery stores were assumed to correlate with lower diabetes rates; however, the finding was that grocery stores increase the diabetes rates. One possible explanation for this result is that counties with higher grocery store density are often smaller, whereas more rural counties would not have as many, as they would be spread thinly. This would mean that the ratio does not correlate directly with easy or equitable food access. However, on the other end, the obesity scatter plot shows a slight negative trend, which is more in line with the idea that grocery stores that are easily accessible are associated with lower obesity; however, this is an inference we can make based on the limited amount of data points. These results show that grocery store density is not a perfect proxy for food access and that future research should find data and factors that contain direct store proximity and food affordability.

Research Question 3: Does the geographical location of each county affect health outcomes and food access?

The choropleth maps of Rural Urban Continuum Codes and grocery store density across U.S. counties show a clear geographic clustering of rural counties and urban counties. Rural counties (higher RUCC codes, shown in the color yellow) are concentrated in the Great Plains, the Mountain West, and parts of the South, while the urban counties are concentrated on the coasts and Midwest corridors. The grocery store density map similarly shows that Alaska and part of the rural West also had a noticeably higher ratio of stores per capita, which goes against the idea of grocery stores and rural areas; however, it reflects the fact that small, remote populations are served by a higher number of small stores relative to the county size. A significant limitation of geographic analysis of three states, Kentucky, Pennsylvania, and Connecticut, were absent on the choropleth visualizations due to missing data, which limits our knowledge and creates blind spots in regions that are potentially important for understanding rural health disparities. This geographic patterning shows the idea that targeting food access and health outcomes would need to be regionally tailored rather than a unified application, as poorer health access appears to be different between the rural South, the Plains, and the West. The maps and the findings reinforce the ideas from the first two research questions by creating a geographical context, showing that health and food access disparities are not random.

Impact and Limitations:

Despite the data sourced from government websites, there are still many counties that didn't have the data we needed in order to make complete visualizations. For example, in the geographical plot of rural-urban continuum codes by county in the United States, 3 states were completely missing from the dataset (Kentucky, Pennsylvania, and Connecticut). When drawing our conclusions, this led to missing values and difficulties extrapolating the results to counties that were missing.

Researchers or policymakers might benefit from our analysis, as it would help to give some insight into patterns between health outcomes and geographical traits. However, since there are many counties and states excluded from our research, it may lead to these places being overlooked or excluded. So further research about the missing values would be ideal.

We also don't intend our results and visualizations to be used to draw any formal conclusions. None of the visualizations is meant to provide hard statistical evidence to prove any specific statistical claim between the variables we explore in this project. The plots are simply intended to explore possible patterns between different variables in the data and shouldn't be used as proof of causation or statistical significance. More formal testing would be needed to make any further claims.

Work Plan:

1. General setup - import data/necessary libraries/get Colab working

Estimated time: (30 minutes)

First, we are going to set up our datasets, libraries, and Colab to work in order to start coding our project. The reason why we chose to use Colab is to ensure participation from both parties in the group with flawless collaboration.

Actual time: (1 hour)

We initially started with working in Google Colab since it's especially good with collaboration and real-time updates. However, since it's a tool that we haven't learned how to use, it took longer to set up and import the datasets than expected.

2. Merging datasets based on the county (2 hours)

Estimated time: (2 hours)

We will then manipulate the data and merge them based on the county, as that is a common column found in all three datasets that we are utilizing for this project. We will use pandas from Python to mainly complete this task.

Actual time: (3 hours)

We had initially planned to merge the data based on county and state names. However, due to the Rural-Urban Continuum dataset having a large variety of additional words added to the county name, like "Municipality", "Borough", and more, we pivoted to using the FIPS as the link between the three datasets. Along with this, we also had to pivot all three of the datasets from a length-wise to a width-wise format, which we had not accounted for when making our initial estimate.

3. Research question 1: Does the level of rural or urban a county is affect the health outcomes of its population?

Estimated time: (3 hours)

Our first research question is to understand if a rural or urban population affects the health outcomes of its population, and we will be manipulating data to allow us to find the connections between the two.

Actual time: (1.5 hours)

After merging all of the data together, most of the work was already completed, so it took less time than expected to make the two boxplot graphs that we needed to answer this question.

4. Research question 2: How does a county's accessibility to food affect the health outcomes of its population?

Expected time: (3 hours)

Similarly to the first research question, we will be finding whether the county's accessibility to food affects the health outcomes of its populations. We will be navigating the data and manipulating it using Pandas to understand and find how a county's accessibility to food increases or decreases the health of its population. We will be creating a scatterplot for this research question.

Actual time: (1.5 hours)

As stated in the previous question, it also took us less time to make the scatterplots necessary to answer this question. So it took around 1.5 hours rather than 3 to complete.

5. Research question 3: Does the geographical location of each county affect health outcomes and food access? (3 hours)

With our last research question, we want to utilize Plotly to help visualize and challenge our project. We are going to find out if the geographical location of each county affects the health outcomes and food access through a data visualization.

Actual time: (3 hours)

Like the last two research questions, since the data was already loaded and merged it was easier. However, this problem did include importing plotly and the necessary geographical county information and learning the basics of choropleth graphs, so it took around the same number of hours as we initially expected.

6. Write report/slides (5 hours)

This is the culmination of our entire project and code, and will have us writing the entire story of our project, showing whether health is impacted by a county's food accessibility, utilizing all three datasets to help find our answer.

Actual time: (5 hours)

This step included splitting up our original code file into two and adding additional files (including a formal testing file and instructions). We spent the rest of the time formalizing our results into this report and creating a presentation of our findings. Overall, it took around the same amount of time to complete as we initially expected.

Work Plan Evaluation:

The work plan that we have proposed has been completed on time, as we have been working on our project throughout the time that we have been spending. We have begun starting the project, creating the Google Colab, and writing all the code for the visualizations and data sets. We have begun merging the datasets to create the visualizations required for the Explanatory Data Analysis checkpoint. The 3 different research questions we had were expected to be done around 3 hours each; however, research question 1 and research question 2 were both done in 1.5 hours, while research question 3 did take the full 3 hours to complete. Our plan was executed to complete the merging process of datasets, which was the largest part of our project, and it took shorter and longer than expected in specific parts of the plan.

Testing:

Regarding testing, we have used a variety of print statements to see the merged datasets as a whole. We compared the merged datasets to ensure that the dataset that we merged for each data was merged correctly. After we see the dataset as correct, we use visualizations to see the merged datasets' results for the regions that have a health impact. We focused on using box plots, scatter plots, and a heatmap in order to find the location of county health impacts. The main testing methods we used were to print the merged datasets to ensure correct merging and to see the results of the health impact.

Collaboration:

The collaboration in this project was only between Sophia Kam and Daniel Kim.